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Aim

To identify the skewness of the distribution shown by each violin plot (made in R with ggplot2; white dot = median) and to state how the mean compares with the median.

How to read a violin plot

Shape of the violin	Skew	Mean vs Median
Wide part (bulk of data) at the bottom, thin tail stretching upwards	Right (positive) skew	Mean > Median
Wide part (bulk of data) at the top, thin tail stretching downwards	Left (negative) skew	Mean < Median
Widest at the middle, both ends tapering evenly; median dot near the centre	Symmetric (no skew)	Mean $\approx$ Median

The tail pulls the mean towards it, while the median stays where most of the data lies. Hence the mean lies on the tail side of the median.

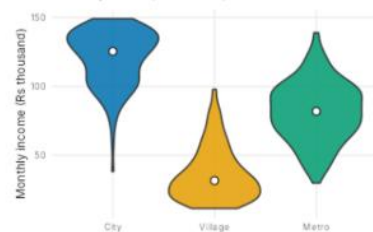
Given charts

DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot C

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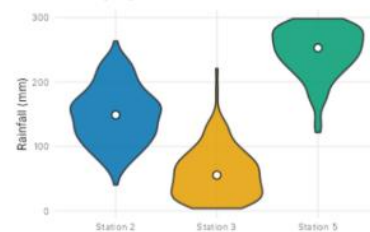
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Monthly income (Rs thousand)



Violin	Skew (circle one)	Mean vs Median (circle)
City	Right / Left / Symm.	Mean > / < / $\approx$ Median
Village	Right / Left / Symm.	Mean > / < / $\approx$ Median
Metro	Right / Left / Symm.	Mean > / < / $\approx$ Median

Chart 2: Rainfall (mm)



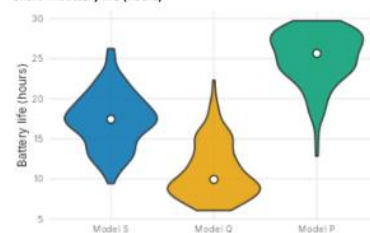
Violin	Skew (circle one)	Mean vs Median (circle)
Station 2	Right / Left / Symm.	Mean > / < / $\approx$ Median
Station 3	Right / Left / Symm.	Mean > / < / $\approx$ Median
Station 5	Right / Left / Symm.	Mean > / < / $\approx$ Median

Chart 3: Sleep per night (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Year 4	Right / Left / Symm.	Mean > / < / $\approx$ Median
Year 1	Right / Left / Symm.	Mean > / < / $\approx$ Median
Year 2	Right / Left / Symm.	Mean > / < / $\approx$ Median

Chart 4: Battery life (hours)



Violin	Skew (circle one)	Mean vs Median (circle)
Model S	Right / Left / Symm.	Mean > / < / $\approx$ Median
Model Q	Right / Left / Symm.	Mean > / < / $\approx$ Median
Model P	Right / Left / Symm.	Mean > / < / $\approx$ Median

Part B (2 marks)

B1. In Chart 2, violin "Station 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical rainfall here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Charts 1-4 as given in the activity sheet.

Part A (18 marks): Skew and Mean vs Median

Chart 1: Monthly income (Rs thousand)

Violin	Skew	Mean vs Median	Reason (from the shape)
City	Left skew	Mean < Median	Violin is widest near the top and the median dot sits high (about 125); a long thin tail runs down to about 40.
Village	Right skew	Mean > Median	Violin is widest at the bottom and the median is low (about 32); a long tail stretches up to about 98.
Metro	Symmetric	Mean $\approx$ Median	Widest around the median (about 82) and tapers about equally above and below; no long one-sided tail.

Chart 2: Rainfall (mm)

Violin	Skew	Mean vs Median	Reason (from the shape)
Station 2	Symmetric	Mean $\approx$ Median	Widest at the median (about 148) with both ends tapering evenly.
Station 3	Right skew	Mean > Median	Bulk of the data at low rainfall (median about 52); a very thin tail extends up to about 220.
Station 5	Left skew	Mean < Median	Bulk of the data at high rainfall (median about 245, near the top); the tail runs down to about 120.

Chart 3: Sleep per night (hours)

Violin	Skew	Mean vs Median	Reason (from the shape)
Year 4	Symmetric	Mean $\approx$ Median	Median (about 6.9) is central in a balanced, diamond-like shape between 5 and 9 hours.
Year 1	Left skew	Mean < Median	Widest near the top with the median high (about 8.9); a thin tail runs down to about 5.8 hours.
Year 2	Right skew	Mean > Median	Widest near the bottom with a low median (about 5); the tail stretches up to about 8 hours.

Chart 4: Battery life (hours)

Violin	Skew	Mean vs Median	Reason (from the shape)
Model S	Symmetric	Mean $\approx$ Median	Median (about 17.5) sits in the middle of an evenly tapering shape from about 9.5 to 26 hours.
Model Q	Right skew	Mean > Median	Bulk at low values with a low median (about 10); a tail extends up to about 22 hours.
Model P	Left skew	Mean < Median	Bulk at high values with the median near the top (about 25.5); the tail runs down to about 13 hours.

Summary of Part A

Chart	Left skew (Mean < Median)	Right skew (Mean > Median)	Symmetric (Mean $\approx$ Median)
1: Income	City	Village	Metro
2: Rainfall	Station 5	Station 3	Station 2
3: Sleep	Year 1	Year 2	Year 4
4: Battery	Model P	Model Q	Model S

Part B (2 marks)

B1. In Chart 2, violin "Station 3" is not symmetric. Towards which side is its tail, and which is the better measure of a typical rainfall here: mean or median? Give one reason. [1]

Answer: The tail of Station 3 is towards the right (the higher rainfall values), so the distribution is right-skewed. The median is the better measure of typical rainfall here. Most rainfall readings are low, and the few very high readings in the long right tail pull the mean upwards, making it larger than a typical value; the median is not affected by these extreme values and stays near where most of the data lies.

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Answer:

```
ggplot(df, aes(x = group, y = value)) + geom_violin() + stat_summary(fun = median, geom = "point", colour = "white")
```

Here geom_violin() draws one violin per group, and stat_summary(fun = median, geom = "point") adds a dot at the median of each group.